ABSTRACT

AETHERIS — Digital Environment Twin System is a web-based application designed to visualize environmental conditions and simplify the monitoring of air quality and microclimate data. It provides an interactive digital representation of environmental sensor stations and displays parameters such as **Air Quality Index (AQI), temperature, humidity, particulate matter, and wind** through a user-friendly dashboard.

The application uses **Java 17, Spring Boot 3.3.5, Spring Data JPA, MySQL, HTML5, CSS3, JavaScript, and HTML5 Canvas**. Its features include user authentication, role-based access control, station management, GPS-based nearest-station identification, live and historical telemetry, and environmental alerts. It also provides a **Clean Air Score (0–100), interactive 2.5D Digital Twin visualization,** hazard simulations, automated telemetry updates, and environmental reports. **The Haversine formula** calculates the distance and bearing to the nearest registered station.

AETHERIS is an **academic prototype** that primarily uses simulated environmental telemetry rather than guaranteed readings from physical sensors. The project demonstrates how web technologies, geospatial calculations, database management, and interactive visualization can make environmental information easy to understand.